



PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

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Report on

**Real-Time IoT-Enabled Water Quality Monitoring and Groundwater
Availability Analysis Using Machine Learning**

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January – May 2024 & August - December 2024

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PROGRAM B. TECH

FACULTY OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

**CERTIFICATE**

This is to certify that the Report entitled
**Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis
Using Machine Learning**

is a Bonafide work carried out by
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In partial fulfillment for the completion of 7th semester course work in the Program of Study B.Tech in Electronics and Communication Engineering under rules and regulations of PES University, Bengaluru during the period Jan – Dec 2024. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of project work.

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DECLARATION

We **Abhishek A Shetty , Akash Chanabasu Katate , Akash Ravi Bhat , Tejas V P** hereby declare that the report entitled, '***Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis Using Machine Learning***', is an original work done by us under the guidance of **Prof. NAGARAJ L J , Assistant Professor , Department of ECE**, is being submitted in partial fulfillment of the requirements for completion of project work in the Program of Study B.Tech in Electronics and Communication Engineering.

PLACE: Bengaluru

DATE: 28/11/2024

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ACKNOWLEDGEMENT

We would like to take this opportunity to thank Prof. NAGARAJ L J, Assistant Professor, Department of Electronics and Communication Engineering, PES University, for his persistent guidance, suggestions, assistance and encouragement throughout the development of this project. It was an honor for me to work on the project under his supervision and understanding the importance of the project at various stages.

We extend our heartfelt thanks to **Dr. J Suryaprasad**, Vice-Chancellor, PES University for his support. We are also grateful to **Dr. Shikha Tripathi**, Chairperson, Department of Electronics and communication, PES University for her support in facilitating the completion of this project.

Finally, we thank our parents, friends and to all the people who were directly or indirectly involved, for their cooperation and support for guiding and supporting us in completion of this project work.

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ABSTRACT

The growing need for pure water emphasizes the necessity of creating innovative solutions that can monitor water quality and provide suitable management strategies for the control of groundwater resources. Our project is "Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis Using Machine Learning." The use of IoT technology and data analysis offers an easy yet potent solution.

Starting with the observation of the trend of historical records on groundwater levels. By using machine learning tools, we detect the patterns and trends in the utilization of groundwater over time. The result of this analysis helps in detecting whether resources are depleting or improving, thus making future decisions to ensure that there will be water availability. IoT sensors are used for monitoring key water quality parameters such as temperature, turbidity, total dissolved solids, and pH levels.

These sensors work continuously and give real-time updates on water quality. Any sudden changes, such as pollution or unusual readings, can be detected and addressed promptly. Our project combines historical groundwater data and real-time water quality readings for an all-inclusive picture of the water situation. The two pieces of data show how related water availability and quality are with each other, and will predict potential risks like shortage or contamination. The findings are presented through graphs, charts, and reports, making the data accessible and easy to understand for everyone, including local communities. Clear visuals enable better planning and proactive measures to protect water resources.

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Chapter 1: Introduction



Fig 1.1: Water Pollution

It has become a growing problem due to water pollution which negatively affects ecosystems, biodiversity, and human health across the world. Polluted water sources not only cause interference in aquatic life but also lead to serious risks against agricultural activities, supplies of drinking water, and general public health.

All this and even more dangers notwithstanding, the conventional methods of monitoring water quality fail to produce the timely data needed to fight pollution incidents effectively. These methods often rely on periodic sampling and laboratory analysis leading to delays in detecting the contamination and responding to potential threats.

This work focuses on the development of an integrated system that uses advanced sensor networks to monitor water quality in real time. Using IoT devices, critical parameters such as pH, turbidity, temperature, and Total Dissolved Solids (TDS) are monitored continuously.

This helps in real-time identification of pollution incidents, and appropriate action can be taken immediately to minimize environmental and health impacts. The project also uses predictive modelling to analyse historical groundwater data and predict future availability trends. Using machine learning techniques, patterns in groundwater depletion or replenishment are identified, providing a clearer understanding of resource dynamics. It integrates real-time water quality monitoring with groundwater predictions into a complete solution for resource water management.



Fig 1.2: Sources of Water Pollution

This approach deals directly with the limitations of the conventional monitoring systems in providing actionable data for stakeholders. The community and policymakers become better empowered by insights that promote quicker, better-informed decisions. In case of a pollution event, for instance, the authorities can immediately implement mitigative measures and thus reduce the long-term damage.

Predictive groundwater information can also lead to sustainable water use practices, which preserve resources for future generations. The rate of pollution incidences is increasing and calls for more than the traditional one. This integrated system with advanced sensor technologies and machine learning offers a proactive and efficient approach to water quality management and estimation of groundwater availability. Such a system is important for resilient communities able to respond to water challenges and protect this resource for years to come.

Chapter 2: Motivation

Water pollution is one of the toughest problems of our time. It threatens environmental health, human well-being, and sustainable development. Industrialization, urbanization, and agricultural activities have increased to such an extent that the pollution level in rivers, lakes, and groundwater sources has reached critical levels. These pollutants damage ecosystems, disrupt biodiversity, and seriously threaten public health by contaminating drinking water and agricultural systems.

This present practice of traditional water quality monitoring is based on sampling and laboratory analysis, usually applied in a periodic manner that leads to delayed response of pollution incidents. This also makes communities open to the shock events and the effects of long-term water degradation. As such, immediate steps cannot be taken toward the protection of water resources due to this openness.

This project views the challenges by enhancing a network of sensors that monitors real-time water quality and the availability of groundwater. Advanced sensor networks continuously measure important parameters, such as turbidity, pH, and TDS, while at the same time integrating history in the development of predictive analysis of the groundwater. The data gap is bridged by existing data, thus providing the stakeholders with information for responding promptly to any pollution event and implementing sustainable water management. Such a project can be no more important. Demand for clean water all over the world is definitely going to increase, and the greatest concern would be that of how clean and safe water could become available. This project thus addresses the immediate

problems of water pollution but also leads towards long-term environmental resilience through smarter, data-driven choices to manage water resources.

Moreover, integrating advanced technologies such as IoT-based sensors with machine learning algorithms enables a more efficient and proactive approach to water management. IoT devices continuously monitor water quality in real-time, providing instantaneous feedback on parameters like dissolved oxygen, electrical conductivity, and chemical pollutants. This data is not only crucial for identifying immediate threats but also feeds into predictive models, helping anticipate future water quality trends and potential crises. Machine learning techniques, such as Gradient Boosting and Random Forest, can process large datasets to uncover patterns and correlations that traditional methods might overlook. By utilizing these insights, stakeholders can implement targeted interventions and policy decisions that prevent irreversible damage to aquatic ecosystems.

In addition to addressing water pollution, this project promotes community engagement and awareness. By visualizing water quality data on accessible platforms like the Blynk IoT dashboard, it empowers citizens to understand and monitor their local water resources. Educational initiatives based on this real-time data can drive behavioral changes at the grassroots level, encouraging water conservation and pollution mitigation. Furthermore, this approach aligns with global sustainability goals, including the United Nations Sustainable Development Goal 6 (Clean Water and Sanitation), by fostering collaborative efforts among governments, industries, and communities to secure water resources for future generations.

Chapter 3: Literature Survey

[1] The author's contribution deals with the correlation analysis between industrial waste discharge and some of the major key parameters, such as pH and temperature, on water quality index. This study offers an IoT-based system of water pollution monitoring in such a way that aims at improving the management of water qualities by alerting concerned authorities in real time. But some limitations exist in the work. Significantly, it is a non-integration system because the machine learning algorithms in advance analysis and predictive models are not available in the system. In such systems, the water quality cannot be evaluated with proper analysis on predicting and making decisions, therefore limited to influence in successful management of water quality.

[2] The author's work deals with a very important issue of water contamination, introducing an IoT-based system for real-time measurement and alerting of water quality. The system monitors the most key parameters like pH, flow, turbidity, and temperature with low-cost sensors. It offers real-time monitoring and has an automated process like tank cleaning and alerting the users for poor quality of water. With a 93% success rate in water problem detection, this system clearly proves its functionality, along with its low cost which makes it affordable. Indicating the scope to further improve the accuracy and durabilities, industrial-grade sensors are specified.

This is one great prospect for future enhancements, bringing out from this study the importance that IoT brings in solving the problems of the world's water quality. Amidst these strengths, though, there are a couple of limitations. The low-cost components usage would degrade the accuracy, and the sensors are sensitive to environmental conditions like

weather. Furthermore, the system is not scalable to meet different standards and also does not consider cybersecurity in the system, which is a very important factor for rolling out the system more. To address these limitations, the project emphasizes the importance of scalability and cybersecurity in its design and implementation.

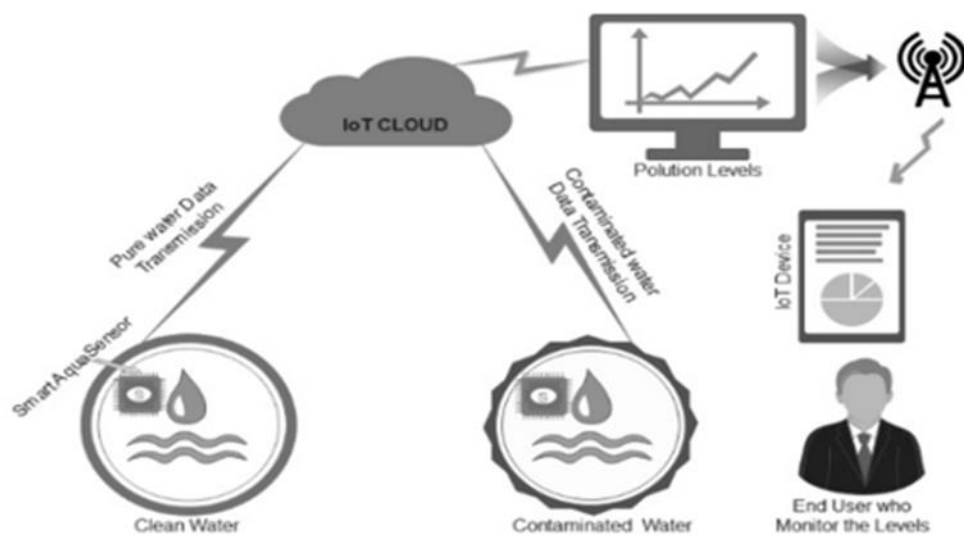


Fig 3.1: Water pollution monitoring with Aqua sensors

By adopting modular architectures, the system can be tailored to meet diverse regional and international water quality standards, making it applicable across various geographic and regulatory contexts. Moreover, integrating robust cybersecurity measures ensures the integrity and confidentiality of data, preventing unauthorized access and potential tampering. This focus on secure and adaptable system design not only builds stakeholder trust but also lays the groundwork for widespread adoption and reliable operation in different environmental conditions and community settings.

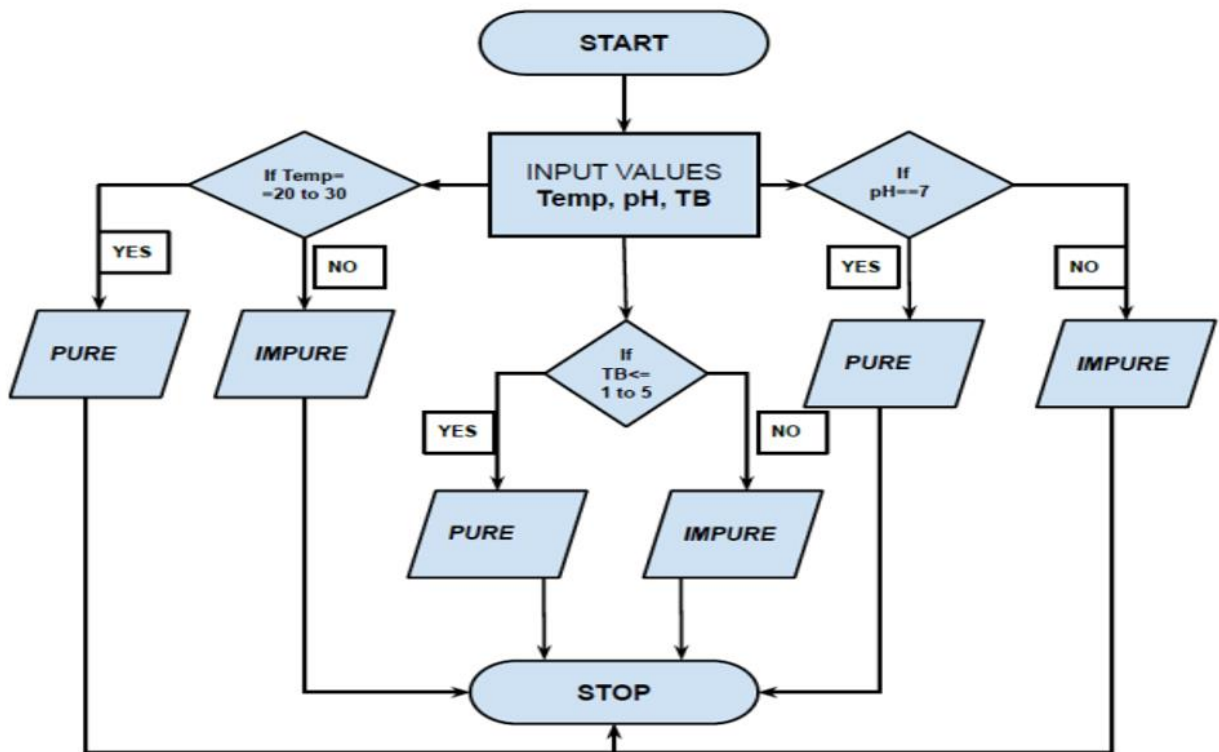


Fig 3.2: System Flow Diagram

[3] The author's work focuses on an IoT-based system designed for real-time water quality and quantity monitoring. The system addresses the challenges of urban water by monitoring key parameters such as turbidity, pH, and temperature. It is designed to be cost-effective and scalable, utilizing low-cost components to make it accessible for wider adoption. The system also holds future potential for expansion, with additional sensors that could support smart city applications. However, this system is less accurate, as it utilizes low-priced components. There are cyber security threats related to data exchange. Therefore, to enhance this system, component quality should be improved along with increasing security

[4] The author's work is a three-layered water monitoring system. This system consists of sensors that collect data, GSM/Ethernet for data transmission, and IoT for processing. EXO Sonde is used to collect data, and monitoring is done on a server. Data is transmitted to the server via GSM, but the real-time data transmission might be affected by connectivity issues. The main limitations of the EXO Sonde are its high

cost and inability to monitor some water quality parameters. The removal of timestamps during data transmission and GSM connectivity problems also results in inconsistent real-time data availability.

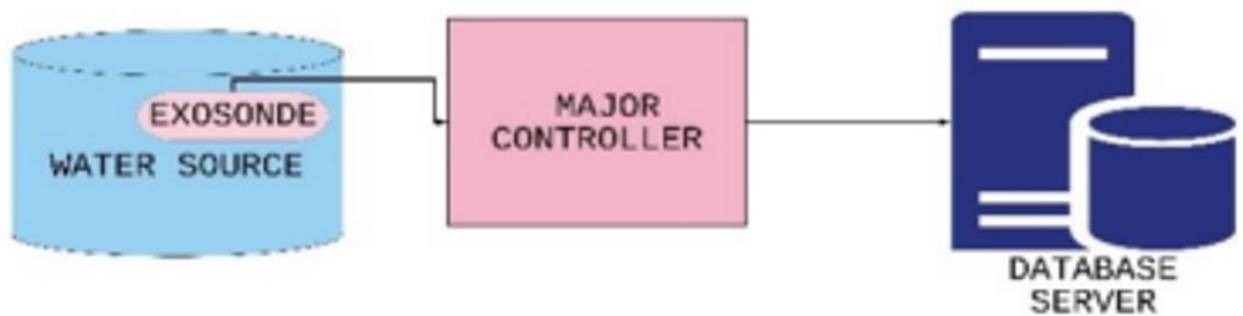


Fig 3.3 : System Overview

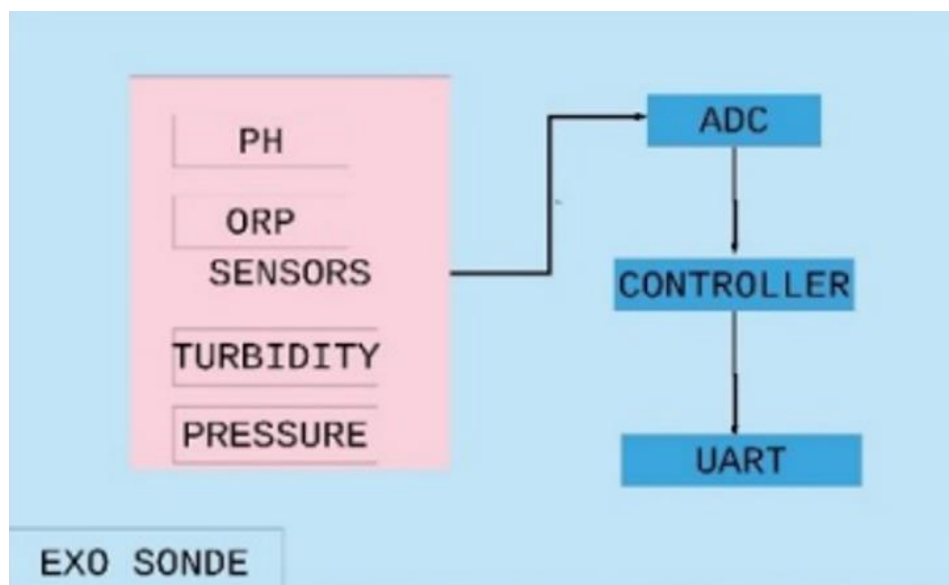


Fig 3.4: Architecture of EXO SONDE

WATER PARAMETER MONITORING				
STN_15_2020017000532_EXO				
Node voltage = 1.5 volts				
PRESSURE (Pascal)	TEMPERATURE (Degree celsius)	PH (no unit)	ORP (milli volt)	TURBIDITY (FTU)
-7841	28.807	9.47	410.05	0.31
-7464	28.812	9.47	410.02	0.30
-7932	28.806	9.47	410.00	0.30
-7971	28.818	9.47	409.97	0.31
-7427	28.815	9.47	409.95	0.27
-7806	28.813	9.47	409.99	0.30
-7865	28.818	9.47	409.86	0.30
-9124	28.817	9.47	409.80	0.28
-8708	28.823	9.47	409.75	0.29
-7683	28.536	9.31	422.29	0.29

Fig 3.5: Monitoring Server

[5] The author's work is about developing an ensemble machine learning model to predict the groundwater level. This would improve the efficiency of borehole drilling and hydraulic resource management. The model uses Boost, AdaBoost, and Gradient Boosting (GB) models to make accurate predictions. Quality data handling is emphasized as a means to enhance prediction accuracy and benefits future studies in the field. However, the limitations of the model include the dependency on the quality of the input data, as the accuracy may be compromised if the data is poor. The model's generalizability might also be limited, as it may not perform well in geographical areas or datasets that were not represented during the model's training.

[6] The author's work is a three-layered water monitoring system. This system consists of sensors that collect data, GSM/Ethernet for data transmission, and IoT for processing. EXO Sonde is used to collect data, and monitoring is done on a server. Data is transmitted to the server via GSM, but the real-time data transmission might be affected by connectivity issues. The main limitations of the EXO Sonde are its high cost and inability to monitor some water quality parameters. The removal of timestamps during data transmission and GSM connectivity problems also result in inconsistent real-time data availability.

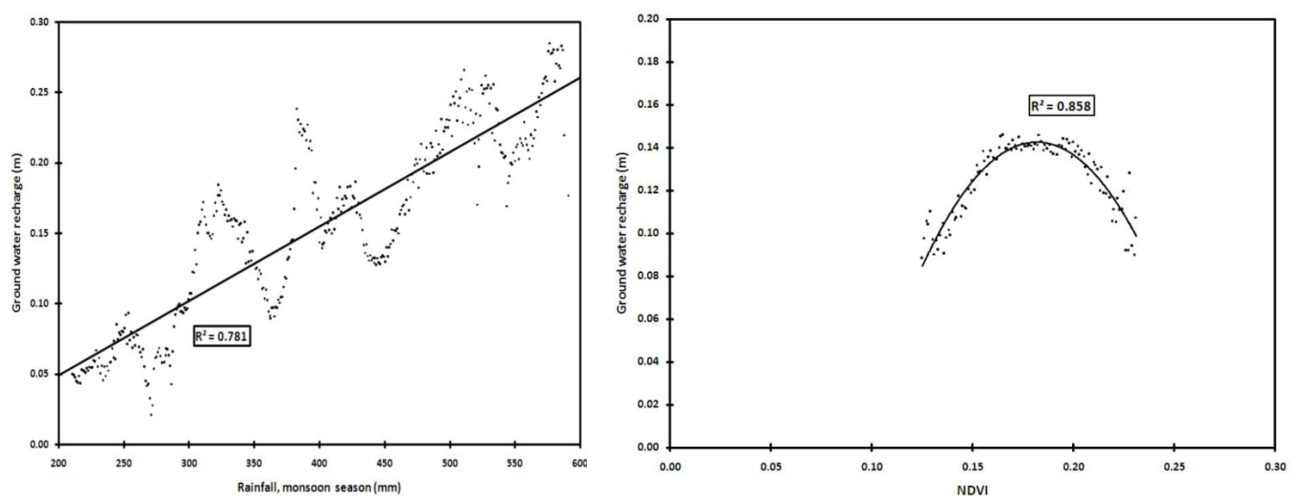


Fig 3.6: Output Graphs

[7] The author works measuring water samples in Bangalore and monitoring parameters such as pH and turbidity. According to the study, temperature is involved in the measurement of Total Dissolved Solids due to its effect on the level of solubility. AQUASENSE used in the system has an advantage of being cost-effective as well as efficient compared to other systems used for monitoring the quality of water. However, the efficiency of the system does rely on the source of water and natural conditions in Bangalore. In addition, calibration can be required periodically to ensure accurate and reliable results.

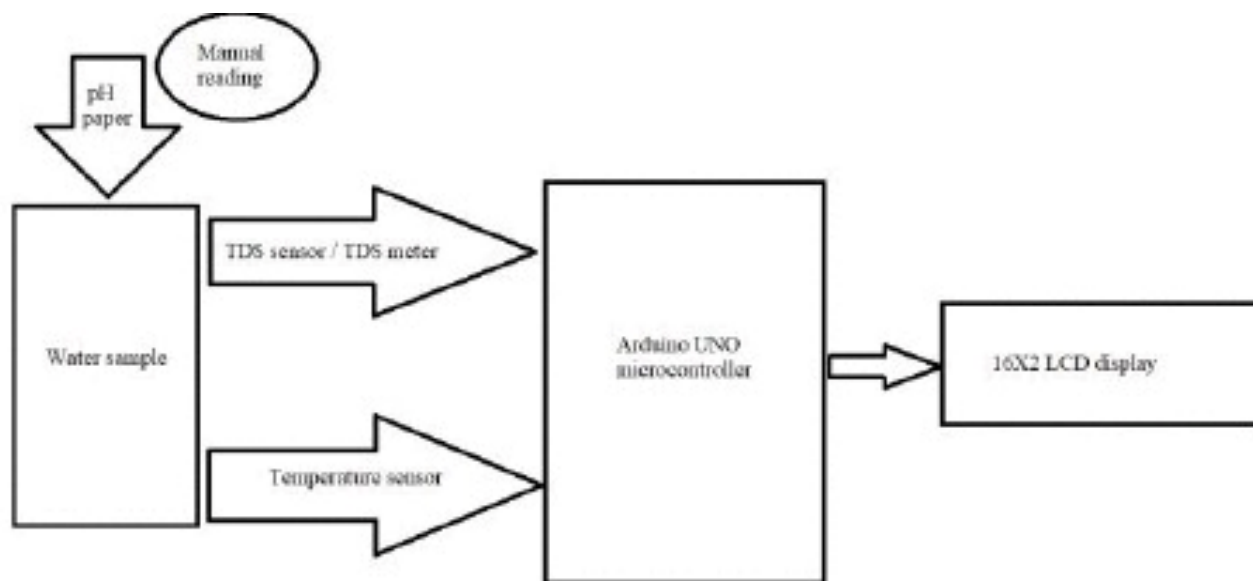


Fig 3.7: Schematic Diagram

[8] The work of the author focuses on the development and application of the SWAT_WAQR model to study water consumption, efficiency, and river water quality in the Yulin Basin. He analyzes the effects of water consumption and pollution on river quality and, as a result, water management practices. He further simulated different scenarios for saving water in Yulin based on forecasting effects that economic usage of water will have, including contamination factors on water quality. Model development assumes and simplifies what it is being constructed and these assumptions may also mean the accuracy of the outcome from using this model is restricted to local conditions and would possibly fail to factor other unrelated influences on water quality, or consumption habits that come along with those outside inputs.

[9] The author work is on predicting the underground water levels by using machine learning and deep learning techniques, focusing more on the superior performance of the Long Short-Term Memory network for time series forecasting. The study shows that LSTM models are the best in providing accuracy in predicting underground water

levels compared to other ML and DL models. However, the work has some limitations. For instance, more diverse data can be included to increase the robustness of the results. Second, the author proposes future avenues by implementing an attention mechanism, further increasing the accuracy of forecasts.

[10] The work of the author is designed in a VM-based model to predict the levels of ground water and consumption of water. It further aims at supporting sustainable resource management. The model analyses different factors, such as geographical data and climatic data for the assessment of water availability toward domestic needs. Using a data-driven approach, the model makes available valuable insights that lead to effective water allocation and particularly in regions where scarcity of water and population growth exist simultaneously. However, the model does have limitations. It is dependent on historical data so that this may limit its predictive accuracy. Its applicability can be restricted by geographical and climatic conditions.

Chapter 4: Methodology

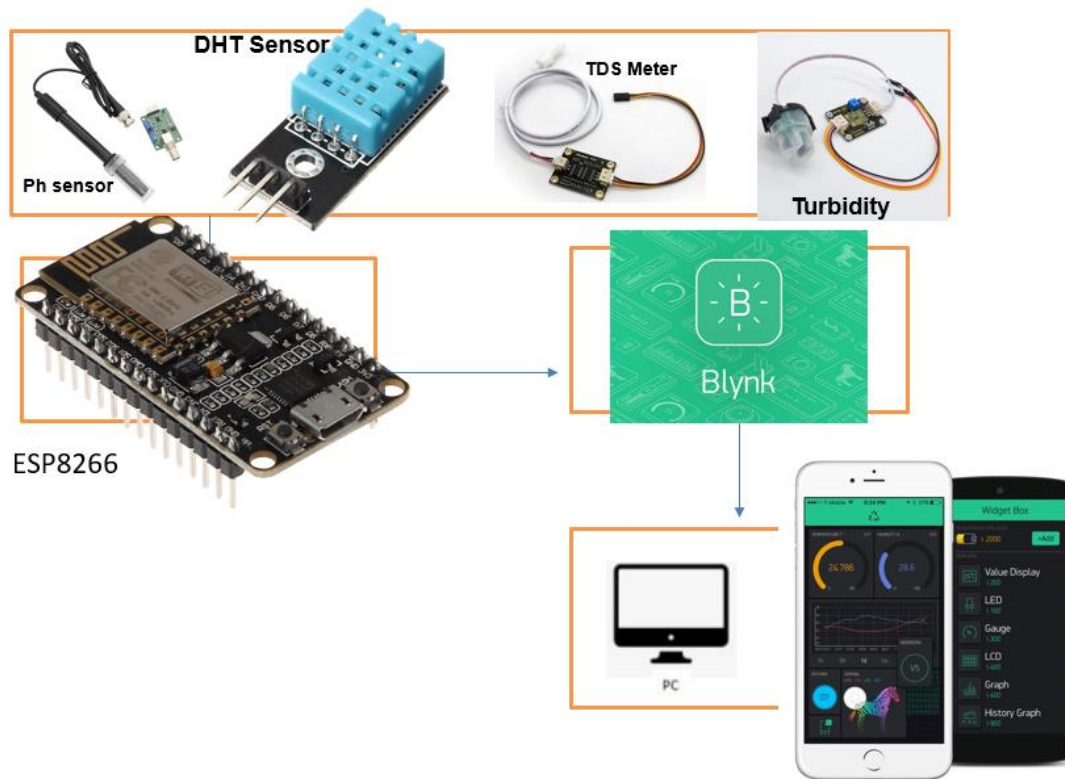


Fig 4.1: water quality monitoring methodology

This methodology involves deploying advanced sensors to measure critical water quality parameters such as pH, turbidity, TDS, and temperature, transmitting the data via a microcontroller to a cloud-based platform. The microcontroller serves as the central processing unit, collecting, preprocessing, and relaying the data in real-time to ensure accuracy and efficiency. The Blynk platform is utilized for visualizing this data, providing an intuitive and interactive interface for users to monitor water quality remotely and in real time. Additionally, alert mechanisms are integrated into the system to notify stakeholders of any deviations from standard water quality levels, enabling prompt action. This setup also allows for data storage and analysis, offering valuable insights for long-term water resource management.

1. pH Sensor

This measures the position of acidity or alkalinity of the water. - DHT Detector Measures temperature and moisture to give an environmental environment to the water quality data. - TDS cadence Measures the Total Dissolved Solids in water, a parameter critical for judging water chastity. - Turbidity Sensor Indicates the clarity of water through measuring suspended patches which indicates implicit pollution.

2. Microcontroller (ESP8266)

The ESP8266 is a microcontroller bedded with Wi- Fi, thus it acts as the central processing unit,

gathers data from detectors and transfers it to the pall or other connected bias over Wi- Fi for farther analysis and visualization.

3. Blynk Application-Blynk is used as the IoT platform for real- time monitoring and control. Detector data is transmitted to the Blynk app, which visualizes parameters like pH, turbidity, TDS, and temperature on a mobile or PC dashboard. - The operation will also enable druggies to define thresholds and admit cautions whenever water quality deviates from the norm.

4. Data Visualization and cautions- The system will display the real- time data on a smartphone or PC through graphs, needles, and other contraptions handed by Blynk. cautions are generated whenever the water quality is poor, enabling druggies to respond consequently.

5. robotization and operations- The system can be set to have automated conduct, for illustration, spark tank cleaning or any corrective measures grounded on set water quality thresholds. This approach illustrates a low- cost effective system to cover real-time water quality, using IoT and stoner-friendly visualization tools to effectively manage water.

This methodology depicts an efficient low-cost system for real-time monitoring of water quality, by utilizing IoT and user-friendly visualization tools for effective management of water.

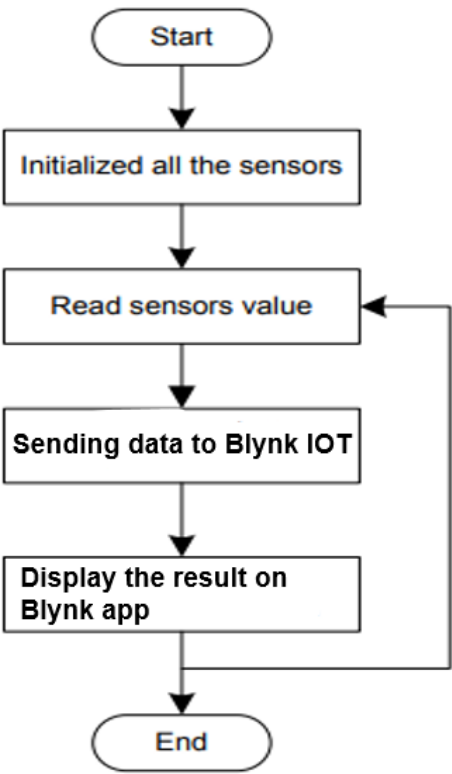


Fig 4.2 Flow Chart of Real time Monitoring Water Pollution

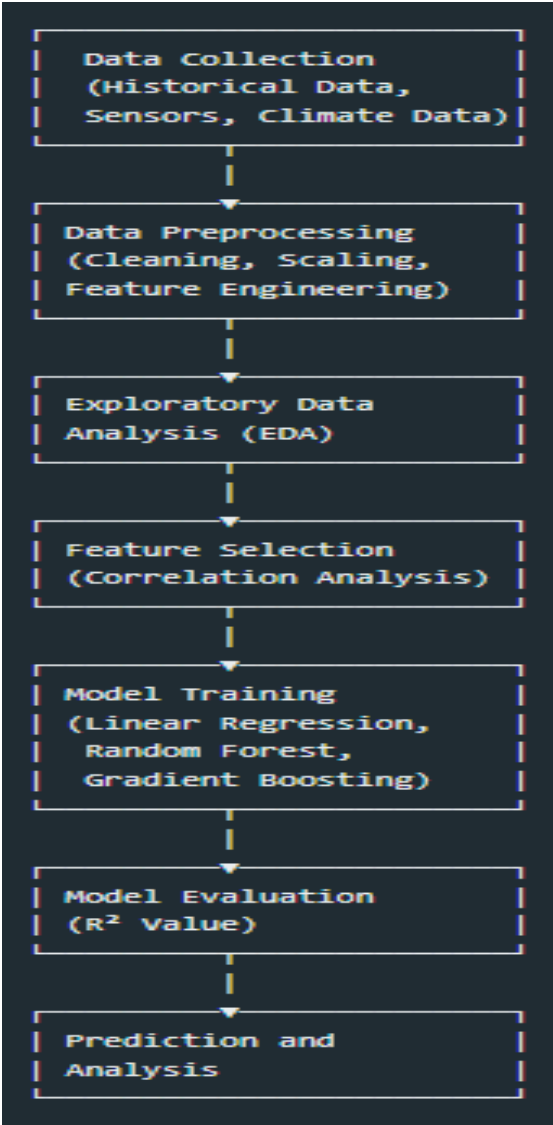


Fig 4.3 Analysis of Ground Water Availability

Chapter 5: Implementation

5.1 Hardware Implementation

- 1. Turbidity Sensor:** Measures water clarity and suspended particles, an important aspect in computing WQI.
- 2. pH Sensor:** Records the acidity or alkalinity of the water, important in deciding water safety.
- 3. TDS Sensor:** Tracks Total Dissolved Solids (TDS), which indicates the mineral concentration in water.
- 4. DHT Sensor:** It measures temperature and humidity, thus analyzing environmental conditions.
- 5. ESP32 Microcontroller:** This is the central component connecting the sensors to the cloud through Wi-Fi for real-time data transmission.

5.2 Software Implementation

1. Blynk IoT Platform:

Data Visualization: A dashboard showing the real-time sensor readings.

Alerts: The platform sends notifications whenever any water parameter exceeds safe thresholds.

2. Python Programming:

Data Cleaning and EDA applied on the India Groundwater Dataset from Kaggle and Data.gov.in to get the data for predictive analysis. ML model: Linear Regression, Random Forest, Gradient Boosting; tried to predict the availability of the groundwater

with the help of previous data.

3. Groundwater Dataset

All the data regarding water availability, level, and quality in India is included within the set.

- EDA provided insights into trends and correlations between water quality parameters.

5.3 Integration Process

- 1. Sensor Data Collection:** Sensors connected to the ESP32 collect real-time readings.
- 2. Cloud Connectivity:** Sensor data is uploaded to the Blynk IoT platform for remote monitoring.
- 3. Machine Learning Predictions:** Groundwater availability predictions are made using pre-trained ML models.

This project brings together a hardware and software product that addresses genuine real-world water management problems, demonstrating scalability, efficiency, and impactful outcomes.

Chapter 6: Result and Analysis

Linear Regression R2 Score: 0.9380242645681784

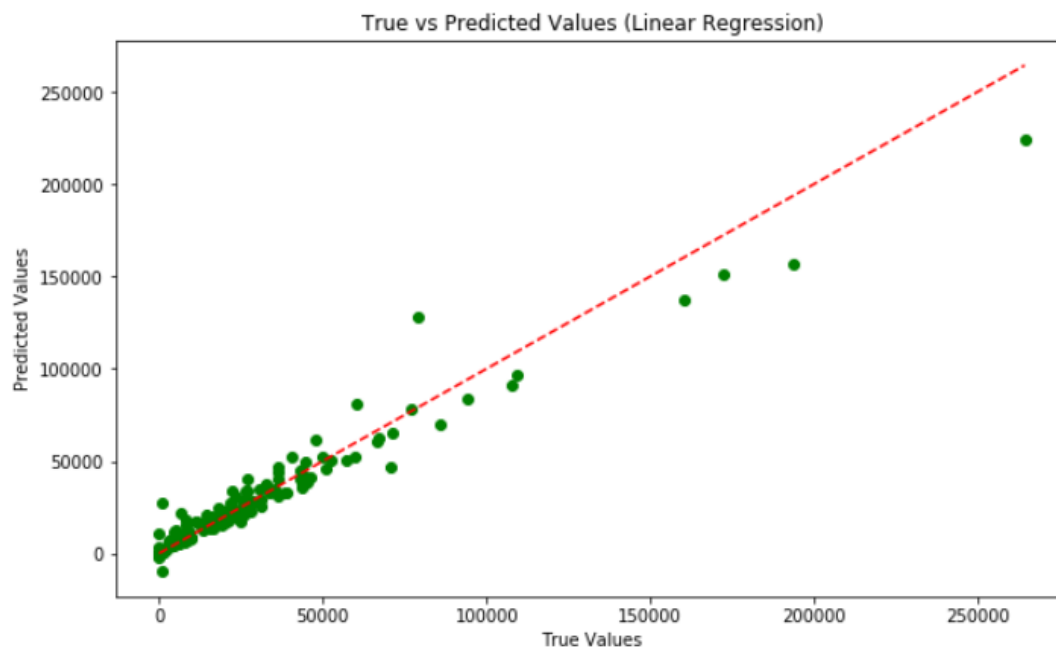


Fig 6.1: Linear Regression Model

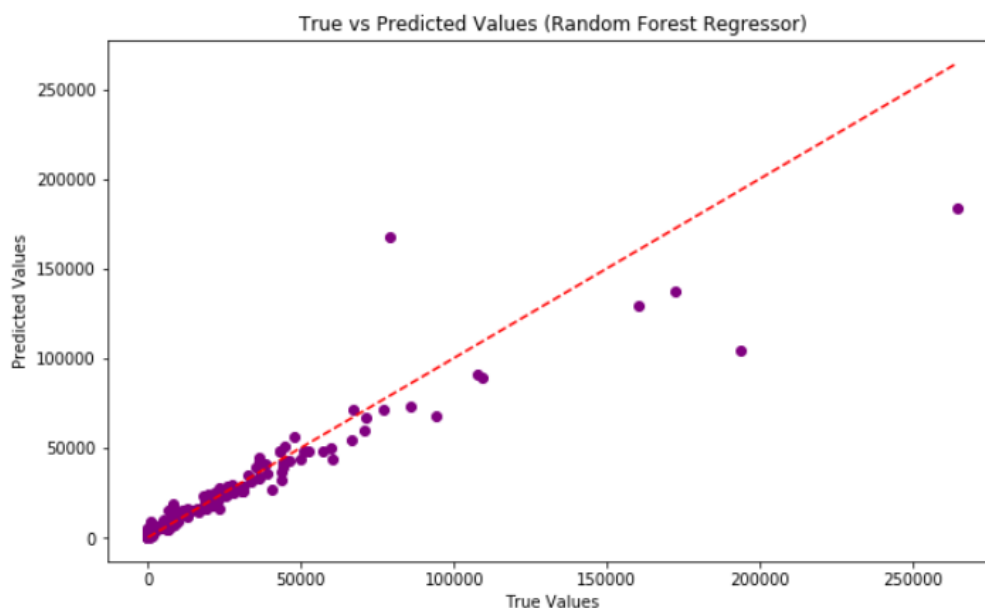
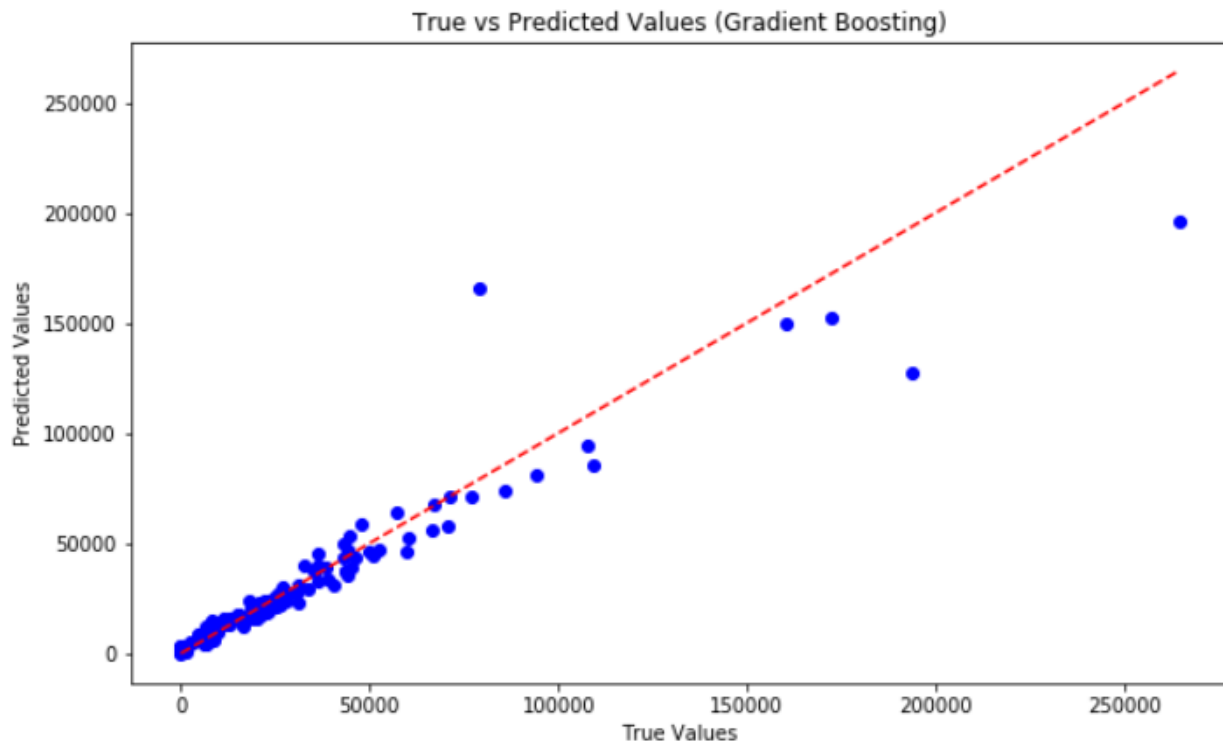


Fig 6.2: Random Forest Regressor

Random Forest Regressor R2 Score: 0.8502557981172599

**Fig 6.3: Gradient Boosting Regressor**

Gradient Boosting Regressor R2 Score: 0.8933798539388684

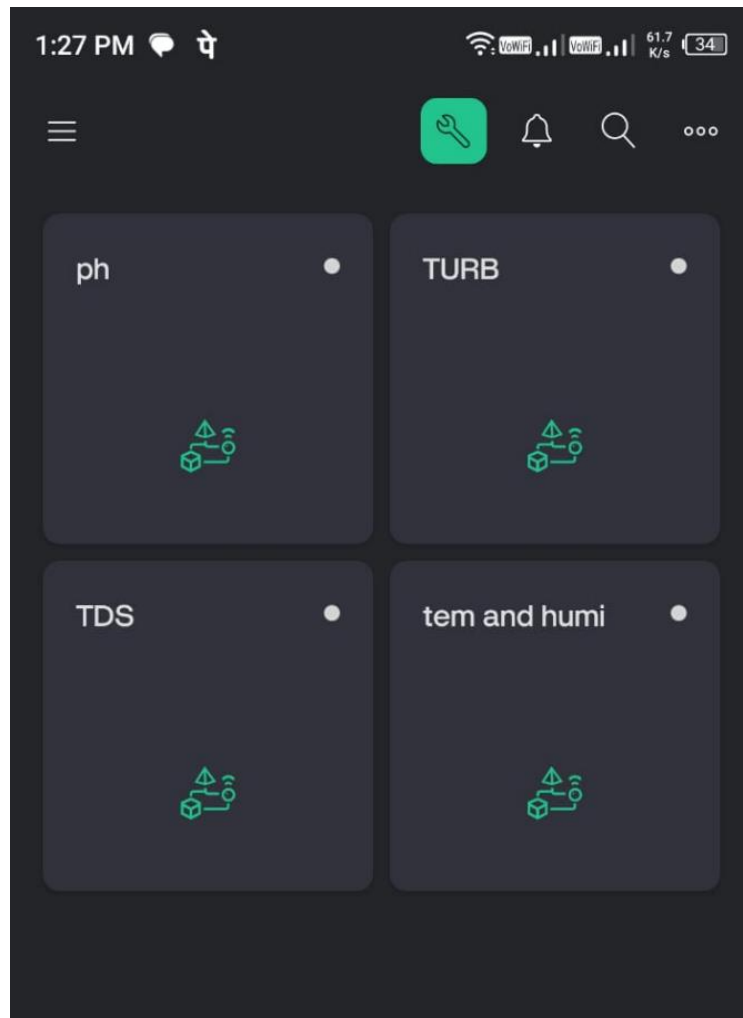


Fig 6.4: Interface of Blynk IOT App

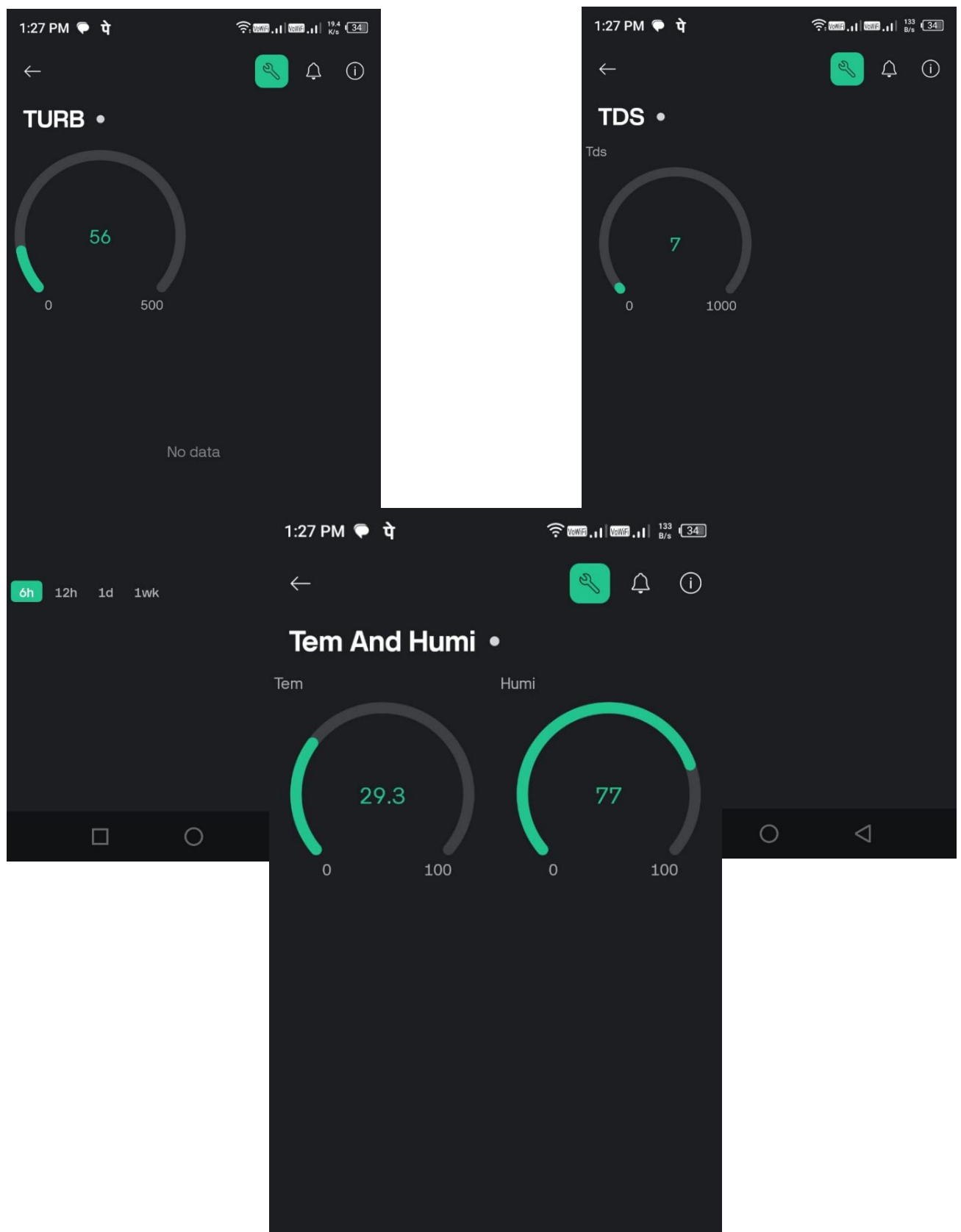


Fig 6.5: Display of sensor output values



Fig 6.6: Circuit connections

Chapter 7: Conclusion

This project will effectively illustrate integration of real-time water quality monitoring through IoT enabled by machine learning predictive analysis on the basis of groundwater availability. The precise and efficient tracking of essential water quality parameters like turbidity, pH, temperature, and TDS will be assured with advanced sensors and cloud-based platforms. The computed Water Quality Index (WQI) is graphically displayed with a user-friendly Blynk IoT dashboard to produce actionable information related to water safety. With these machine learning techniques applied to historical data, predictions regarding the availability of groundwater were accurate enough to greatly aid in sustainable water resource management.

The project also indicates the live data importance for timely pollution event detection, identification of sources of contamination, and targeted interventions. It further emphasizes the ability of predictive analytics to understand groundwater dynamics and address climate change as well as growing water demand-related challenges.

By seamlessly integrating IoT-based real-time monitoring with predictive analytics, this project ensures a dynamic and adaptive approach to water resource management. Advanced sensors transmit continuous data streams to the cloud, facilitating remote access and real-time visualization for all stakeholders. This enables rapid decision-making during pollution events and offers a comprehensive understanding of the trends affecting water bodies over time. Additionally, the use of cloud platforms ensures scalability, making the system adaptable for larger geographic regions and varying water management needs.

The inclusion of machine learning in this project strengthens its predictive capabilities by analysing patterns in historical and real-time data to forecast groundwater availability and potential pollution events. Models like Linear Regression, Random Forest, and Gradient Boosting are utilized to evaluate parameters such as rainfall patterns, seasonal variations, and aquifer recharge rates. These predictive insights allow for proactive measures, such as optimizing water usage, controlling industrial discharge, and planning efficient irrigation systems, ensuring long-term sustainability of water resources.

This project not only tackles immediate water quality and availability challenges but also contributes to environmental resilience and resource optimization in the face of climate change. By creating a robust framework for monitoring and analysis, it equips communities, industries, and policymakers with the tools to safeguard water resources. Furthermore, it paves the way for further advancements, such as integrating satellite data for broader spatial analysis or employing AI-driven models for enhanced prediction accuracy, solidifying its role as a cornerstone for sustainable water management.

Chapter 8: Future Directions

8.1 Improved Monitoring Capability

More Extensive Parameter Monitoring: I would include measurements of heavy metal contamination, nitrates, and biological pollutants in the IoT system to gain a better insight into water quality.

Superior Sensors: Upgrading to high-end sensors will further improve the accuracy and reliability.

8.2 Advanced Predictive Models

Deep Learning and Time-Series Analysis: Incorporation of deep learning models and time-series analysis will strengthen predictions about groundwater availability.

Ensemble Methods: Ensemble techniques will be utilized in order to increase the reliability and precision of the predictions, thereby using multiple machine learning algorithms.

8.3 Policy Integration

Partnership with Governmental Authorities: In this step, the system would be integrated into the national frameworks on water management in collaboration with policy-makers.

Actionable Insights: Meaningful insights can assist in formulating effective water conservation and pollution control measures.

8.4 Scalability and Community Engagement

Wider Rollout: Scaling the project to roll out in various regions will address local and global water issues.

Mobile Application: A mobile application will increase community involvement by making available real-time data and information to local stakeholders.

8.5 Sustainability Emphasis

Renewable Energy Integration: Researching and incorporating renewable energy sources, such as solar power, will ensure the development of a green and energy-efficient IoT system.

Equitable Resource Delivery: It will be optimized to serve rural and underrepresented communities for the delivery of water monitoring and management solutions.

Chapter 9: References

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Chapter 10: Appendices

10.1 pH Sensor

The pH detector is one of the tools that measure the acidity or alkalinity of water, taken in a scale from 0 to 14. A logarithmic scale assesses the position of acidity or stipulation in a water sample. However, also the water is acidic, while if it's lesser than 7, If the pH is lower than 7. dimension is important for water quality analysis since pH is one of the factors that affect chemical and natural processes in water bodies.

For case, extreme pH values affect submarine life, alter the solubility of heavy essence, and affect the effectiveness of disinfection processes. Monitoring similar conditions is veritably important in environmental and artificial operations where balanced pH conditions must be maintained.

10.2 TDS Detector

The graveness Analog TDS Detector measures the attention of Total Dissolved Solids (TDS) in water, with a measuring range of 0 to 1000 corridor per million(ppm) and delicacy of ± 10 at 25 °C. The TDS is one of the crucial pointers of the quality of water because indeed small situations can indicate dangerous pollutants, similar as arsenic, iron, manganese, sulfate, and platitude.

High TDS attention not only affect in health hazards but also affect the taste, odor, and delectability of water. Monitoring TDS is thus essential in pollution control, water sanctification processes, and felicity for drinking and artificial purposes.

10.3 Turbidity Sensor

A turbidity detector measures the cloudiness or haziness of a liquid with a seeing capacity range of 5 to 4000 NTU (Nephelometric Turbidity Units). These detectors have great applicability to the assessment of water quality, substantially in the use with veritably high attention of Total Suspended Solids and TDS.

Light intensity variations, when detected by the detector, may be employed for determining particulate matters within the water; still, proper external lighting hindrance is anticipated at dimension time to give valid data. Turbidity monitoring is used considerably in artificial processes, wastewater treatment, and environmental studies to minimize waste, optimize operations, and assess water quality.

10.4 DHT Detector

The DHT detector measures temperature and moisture. The moisture seeing range is 20 to 80 RH, and the temperature seeing range is 0 °C to 50 °C, depending on the model. It uses a capacitive moisture seeing element and a thermistor for temperature seeing.

DHT detectors are primarily used in environmental monitoring, heating, ventilation, and cooling systems, and husbandry sectors for maintaining optimal conditions using climate parameter shadowing. For operations that bear accurate and precise monitoring of atmospheric conditions to ensure functional effectiveness and sustainability, these detectors are one of the most precious tools of the trade.

10.5 Linear Regression

Linear Regression is a core machine learning model that shows how the target variable relates to the input features linearly. In this project, it was used on historical data related to ground water availability. It is simple to implement and interpret and a good

choice for any baseline model for comparison.

While it works well on datasets that are linear in trend, its simplicity means it cannot capture complex, non-linear relationships that often exist in groundwater availability patterns. Even so, the model was useful for understanding general trends and as a benchmark to judge the performance of more advanced algorithms.

10.6 Random Forest Regression

Random Forest is an ensemble learning method; hence, it uses the power of multiple decision trees to improve the accuracy of the prediction, thus minimizing overfitting. For this project, the Random Forest Regression effectively caught the non-linear relationship as well as interactions between variables related to groundwater.

With the average of numerous predictions through decision trees, the model achieved robust performance and noise resilience in the data. The feature importance analysis by the algorithm provided insights into what factors are important in affecting groundwater availability, thereby increasing interpretability. However, its computational complexity was relatively higher, especially with large datasets, requiring careful parameter tuning for optimal results

10.7 Gradient Boosting Algorithm

Gradient Boosting is a very powerful ensemble technique that builds predictive models sequentially, where each tree corrects the errors of its predecessors. With iterative learning, it optimized the performance and was very useful in predicting groundwater availability.

Other models were outperformed, but Gradient Boosting still could not handle complex relationships between variables and provided high accuracy in this project. Still, it needed the tuning of hyperparameters, for instance, learning rate, the number of estimators, and tree depth in order to avoid overfitting and improve generalization.

Despite the longer training time, the algorithm has shown its strength in terms of predictive performance and handling imbalanced data, making it the preferred choice for real-world applications in water resource management.

10.8 Blynk IOT

Blynk IoT is a versatile platform designed for building and managing Internet of Things (IoT) applications with ease. It provides a user-friendly interface for visualizing real-time data, enabling remote control and monitoring of connected devices. With its customizable dashboards, users can integrate various widgets like graphs, buttons, and gauges to represent data interactively.

Blynk supports multiple communication protocols such as Wi-Fi, Bluetooth, and GSM, making it adaptable for diverse IoT projects. Additionally, its cloud-based infrastructure ensures reliable data storage and accessibility from anywhere, while features like notifications and automated alerts enhance functionality, making it ideal for applications like water quality monitoring, smart home systems, and industrial automation.

Codes:

PH SENSOR

```
const int pHSensorPin = A0; // Pin connected to pH sensor output

const float slope = 3.5;    // Calibration slope (adjust based on your sensor)

const float offset = 0.0;   // Calibration offset (adjust based on your sensor)

const float VREF = 5.0;    // Reference voltage for ADC (usually 5V)

// Variables to hold sensor data
```

```
float pHValue = 0.0;    // pH value calculated from sensor

// Calibration constants (you may need to adjust based on your specific sensor and
calibration)

const float calibrationLow = 4.0; // pH value of the solution with known pH (e.g.,
pH 4)

const float calibrationHigh = 7.0; // pH value of another known solution (e.g., pH 7)

const int calibrationLowADC = 200; // ADC value for calibration low (corresponds
to pH 4)

const int calibrationHighADC = 600; // ADC value for calibration high (corresponds
to pH 7)

// Serial output delay

const int delayTime = 1000; // Delay time between readings (1 second)

void setup() {

    // Start serial communication

    Serial.begin(9600);

    // Wait for the serial connection to establish (useful for some boards like Arduino
Leonardo)

    while (!Serial);

    // Print startup message
```

```
Serial.println("pH Sensor Calibration and Monitoring System");

Serial.println("Calibrating pH sensor...");

delay(2000); // Wait for the user to read the instructions
}

void loop() {

    // Read the analog value from the pH sensor

    int sensorValue = analogRead(pHSensorPin);

    // Convert the sensor value to pH value directly using the calibration function

    pHValue = calculatePH(sensorValue);

    // Ensure the pH value is within the expected range (0-14)

    pHValue = constrain(pHValue, 0.0, 14.0);

    // Display the pH value

    Serial.print("pH: ");

    Serial.println(pHValue, 2); // Print pH value with 2 decimal places

    // Delay before next reading

    delay(delayTime);

}
```

```
// Function to calculate the pH value from the ADC reading

float calculatePH(int sensorValue) {

    // Convert the ADC reading to voltage

    float voltage = sensorValue * (VREF / 1023.0); // 10-bit ADC (1024 levels)
    mapped to 0-5V

    // Linear calibration based on two points: low (e.g., pH 4) and high (e.g., pH 7)

    // You may need to adjust the slope and offset values based on your specific sensor
    calibration

    float slope = (calibrationHigh - calibrationLow) / (calibrationHighADC -
    calibrationLowADC);

    float offset = calibrationHigh - (slope * calibrationHighADC);

    // Calculate the pH value from the voltage

    int adcReading = voltage * 1023.0 / VREF; // Convert voltage back to ADC
    reading (between 0 and 1023)

    float pH = slope * adcReading + offset; // Apply calibration equation to calculate
    pH

    return pH;

}
```

Turbidity Sensor code:

```
#define BLYNK_TEMPLATE_ID "TMPL3mWfugSm-"
```

```
#define BLYNK_TEMPLATE_NAME "TURB"

#define BLYNK_AUTH_TOKEN "mCOLfunVMhaFI49ZpyHtFKnedf8OKx-S"

#define BLYNK_PRINT Serial


#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>


char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "Samir"; // Your Wi-Fi name

char pass[] = "Samir123"; // Your Wi-Fi password


BlynkTimer timer;


// Analog pin for turbidity sensor

#define TURBIDITY_PIN A0 // Connect the turbidity sensor output to A0


// Function to read turbidity and scale the value

int readTurbidity() {

    int rawValue = analogRead(TURBIDITY_PIN); // Read the sensor value

    int scaledValue = map(rawValue, 0, 1023, 100, 0); // Scale to 0-100

    return constrain(scaledValue, 0, 100); // Ensure it's within 0-100

}
```

```
// Function to send sensor data to Blynk

void sendSensorData() {

    int turbidityValue = readTurbidity(); // Get the turbidity reading


    // Print turbidity value to Serial Monitor

    Serial.print("Turbidity Value: ");

    Serial.println(turbidityValue);


    // Send scaled turbidity value to Blynk

    Blynk.virtualWrite(V0, turbidityValue); // V0 is the virtual pin for turbidity
}


void setup() {

    Serial.begin(115200);

    Blynk.begin(auth, ssid, pass);


    // Set up timer to send sensor data every second

    timer.setInterval(1000L, sendSensorData);

}


void loop() {

    Blynk.run();

    timer.run();

}
```

```
}
```

TDS Sensor code:

```
#define BLYNK_TEMPLATE_ID "TMPL3uljaezBV"

#define BLYNK_TEMPLATE_NAME "TDS"

#define BLYNK_AUTH_TOKEN "8swe0FaLy4xOzs2WtBlhW6gOsANPqoBw"

#define BLYNK_PRINT Serial


#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>


char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "Samir"; // Your Wi-Fi name

char pass[] = "Samir123"; // Your Wi-Fi password


BlynkTimer timer;


// Analog pin for TDS sensor

#define TDS_PIN A0 // Connect the TDS sensor output to A0


void sendSensor() {

    int tdsValue = analogRead(TDS_PIN); // Read the TDS sensor value
```

```
// Print TDS value to Serial Monitor

Serial.print("TDS Value: ");

Serial.println(tdsValue);


// Send TDS value to Blynk

Blynk.virtualWrite(V0, tdsValue); // V0 is the virtual pin for TDS

}


void setup() {

  Serial.begin(115200);

  Blynk.begin(auth, ssid, pass);


// Set up timer to send sensor data every second

timer.setInterval(1000L, sendSensor);

}


void loop() {

  Blynk.run();

  timer.run();

}
```

DHT Sensor

```
#define BLYNK_TEMPLATE_ID "TMPL3oKyonPk7"
```



```
#define BLYNK_TEMPLATE_NAME "tem and humi"

#define BLYNK_AUTH_TOKEN "OT3juyza8uWH6PeMILCF_u-MfmtGddSs"

#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>


#include <DHT.h>


char auth[] = BLYNK_AUTH_TOKEN;


char ssid[] = "Samir"; // type your wifi name
char pass[] = "Samir123"; // type your wifi password


BlynkTimer timer;


#define DHTPIN 2 //Connect Out pin to D4 in NODE MCU

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);
```

```
void sendSensor()

{

    float h = dht.readHumidity();

    float t = dht.readTemperature(); // or dht.readTemperature(true) for Fahrenheit


    if (isnan(h) || isnan(t)) {

        Serial.println("Failed to read from DHT sensor!");

        return;

    }

    // You can send any value at any time.

    // Please don't send more that 10 values per second.

    Blynk.virtualWrite(V0, t);

    Blynk.virtualWrite(V1, h);

    Serial.print("Temperature : ");

    Serial.print(t);

    Serial.print("  Humidity : ");

    Serial.println(h);

}

void setup()

{

    Serial.begin(115200);
```

```
Blynk.begin(auth, ssid, pass);  
  
dht.begin();  
  
timer.setInterval(100L, sendSensor);  
  
}  
  
void loop()  
{  
  
  Blynk.run();  
  
  timer.run();  
  
}
```

Poster

Real-Time IoT-Enabled Water Quality Monitoring and Groundwater Availability Analysis Using Machine Learning

Abhishek A Shetty , Akash Chanabasu Katate , Akash Ravi Bhat , Tejas V P

Project guide : Prof. NAGARAJ L J



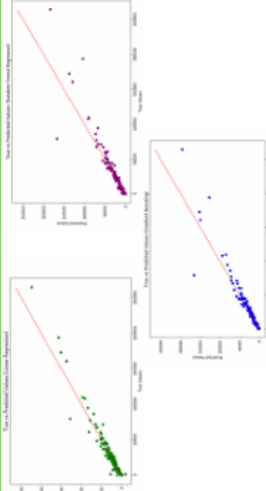
Motivation

This motivation highlights the urgency of addressing water pollution, which poses significant threats to environmental health, public well-being, and sustainable development. With rising pollution levels driven by industrialization, urbanization, and agriculture, traditional monitoring methods are inadequate for providing real-time data. This project addresses the critical need for continuous monitoring, enabling rapid responses to pollution and informed decision-making to protect water resources.

Theory

Groundwater availability is crucial for sustainable resource management, requiring a combination of historical data analysis, real-time monitoring. Machine learning models can analyze trends and forecast groundwater levels, while IoT sensors continuously track water quality parameters like pH, turbidity, and TDS. By integrating these technologies and visualizing results through dashboards and reports, decision-makers can effectively manage groundwater resources, mitigate over-extraction risks, and ensure water quality for various uses.

Analysis



Implementation

Hardware Implementation

The project uses sensors like turbidity, pH, TDS, and DHT to monitor water quality and environmental conditions. These sensors are crucial for measuring clarity, acidity, mineral concentration, temperature, and humidity.

An ESP32 microcontroller integrates these sensors and connects to the cloud via Wi-Fi. This setup enables real-time data transmission and ensures efficient remote monitoring. The system is designed to operate reliably in diverse environmental conditions, ensuring consistent performance.

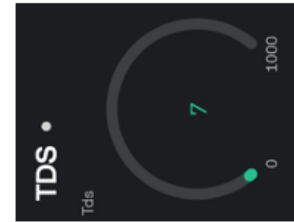
Software Implementation

The Blynk IoT platform provides real-time visualization of sensor data and sends alerts for unsafe water parameters. It ensures accessible and actionable data for stakeholders.

Python programming is used for data cleaning, analysis, and machine learning on groundwater datasets. Models like Linear Regression and Random Forest predict groundwater availability for sustainable management.

Results and Conclusion

This project will effectively illustrate integration of real-time water quality monitoring through IoT enabled by machine learning predictive analysis on the basis of groundwater availability. The precise and efficient tracking of essential water quality parameters like turbidity, pH, temperature, and TDS will be assured with advanced sensors and cloud-based platforms.. With these machine learning techniques applied to historical data, predictions regarding the availability of groundwater were accurate enough to greatly aid in sustainable water resource management.



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